8.1 After Filters and Around Filters

In addition to "before" filters, you can also run filters after an action has been executed, or both before and after.

"After" filters are similar to "before" filters, but because the action has already been run they have access to the response data that's about to be sent to the client. Obviously, "after" filters cannot stop the action from running.

"Around" filters are responsible for running their associated actions by yielding, similar to how Rack middlewares work.

For example, in a website where changes have an approval workflow an administrator could be able to preview them easily, just apply them within a transaction:

```
class ChangesController < ApplicationController
  around_action :wrap_in_transaction, only: :show

  private

  def wrap_in_transaction
    ActiveRecord::Base.transaction do
      begin
        yield
      ensure
        raise ActiveRecord::Rollback
      end
    end
  end
end
```

Note that an "around" filter also wraps rendering. In particular, if in the example above, the view itself reads from the database (e.g. via a scope), it will do so within the transaction and thus present the data to preview.

You can choose not to yield and build the response yourself, in which case the action will not be run.

8.2 Other Ways to Use Filters

While the most common way to use filters is by creating private methods and using *_action to add them, there are two other ways to do the same thing.

The first is to use a block directly with the *_action methods. The block receives the controller as an argument, and the `require_login` filter from above could be rewritten to use a block:

```
class ApplicationController < ActionController::Base
  before_action do |controller|
    unless controller.send(:logged_in?)
      flash[:error] = "You must be logged in to access this section"
      redirect_to new_login_url
    end
  end
end
```

Note that the filter in this case uses `send` because the `logged_in?` method is private and the filter is not run in the scope of the controller. This is not the recommended way to implement this particular filter, but in more simple cases it might be useful.